

SECTION

ADVANCED COMPUTER VISION

CHANGE DETECTION • SALIENCY DETECTION • ANOMALY DETECTION

SECTION

ADVANCED COMPUTER VISION

CHANGE DETECTION



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

What Is Change Detection?

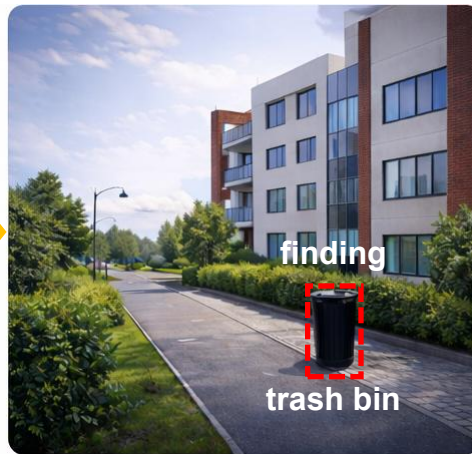
In Computer Vision tasks, **Change Detection** is the process of **finding and mapping** what has changed in a scene by comparing a **reference frame** (i.e., the baseline image **at time t_1**) with a **current frame** (i.e., the newly acquired image **at time t_2**) of the same area. In practice, the goal is to produce a **change map**, often binary (changed/unchanged) or multi-class (type of change), that highlights **where relevant differences** occurred between the two (or more) acquisitions.

Reference Image



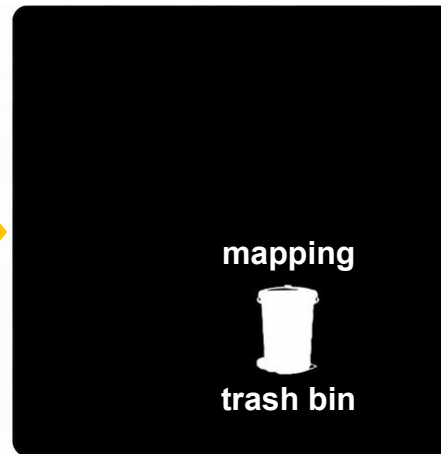
Scene
Before Change

Current Image



Scene
After Change

Change Map



Detected
Change

Notes:

Note (1): A **Change Map** (e.g., binary change map in this example) is an **output image** where **each pixel** indicates whether a change occurred **at that location** (x,y):

- $CM(x,y) = 0$ (Black) $\rightarrow$ Unchanged;
- $CM(x,y) = 1$ (White) $\rightarrow$ Changed.

Note (2): **Change Maps** can take different forms (e.g., binary, probabilistic, multi-class), but the most basic and widely used one is the binary mask (changed vs unchanged).



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

Key Question: What Do We Mean by “Change”?

In Computer Vision, a **change** can be interpreted as **any modification of pixel** values observed in the **current frame** with respect to a **fixed reference frame**. Such **pixel-level modifications** may be caused by:

Appearance of New Objects or Regions



Modification of Object or Regions Properties (e.g., Texture)



Disappearance of Existing Objects or Regions



Movement or Displacement of Objects or Regions



Note:

It is important to observe that each new image acquired at a later time is always compared to the **same reference frame** (baseline at time t_1 , at least in basic and not overly complex scenarios).

In other words: relative to the **reference frame**, any event that causes a **relevant change** in a pixel region in the **current frame**, e.g. by the appearance, disappearance, modification, or movement/displacement of scene elements, can be considered a change.



ADVANCED COMPUTER VISION

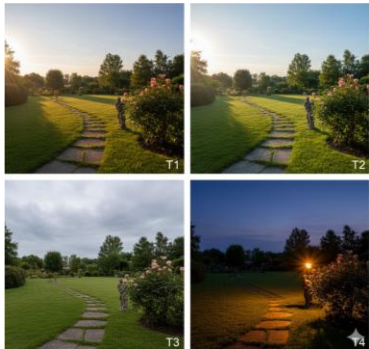
Change Detection • Saliency Detection • Anomaly Detection

Key Question: About the “Relevant Change in a Pixel Region”

Note that, even in a **very simple setting** (i.e., fixed camera, controlled environment), if we capture one image per second without any scene change (i.e., **without** appearance of new objects or regions, **without** disappearance of existing objects or regions, **without** modification of object or regions properties, or **without** movement/displacement of objects or regions), **we should not expect two frames to be pixel-wise identical**. These differences may be due to several factors:



(A) – First Class – Acquisition-Related Variability (Cause → Effect):



A.1 – Illumination Variability:

In real acquisitions, the lighting conditions of a scene are not perfectly stable over time, even when nothing in the scene actually changes. As a result, **the same surfaces may appear brighter or darker**, and shadows may slightly shift, thus producing apparent changes in pixel values.

- (Cause): e.g., sun position, indoor lights flicker, small shadow shifts, clouds → (Effect): global/local intensity changes.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection



A.2 – Sensor Noise and Sensitivity Drift:

Even when the scene is perfectly static, **the camera sensor does not measure light in a perfectly stable way**. Random electronic fluctuations and small variations in the acquisition chain **introduce noise, causing pixel values to slightly change from frame to frame**.

- (Cause): sensor noise & sensitivity drift (e.g., shot/thermal noise, gain instability) → (Effect): random pixel fluctuations.



A.3 – Tiny Micro-Motions:

Even with a “fixed” camera, **tiny vibrations can slightly shift the image over time**. These shifts are often sub-pixel, so the scene is sampled at slightly different pixel locations across frames. As a result, **pixel-by-pixel comparisons will show differences**, even when the scene is unchanged.

- (Cause): e.g., vibrations, micro-shake → (Effect): sub-pixel misalignment (“false” pixel differences (especially near edges)).

Final Notes About: Acquisition-Related Variability (First Class):

Note (1): Even with a “fixed” camera, **illumination variability, sensor noise/sensitivity drift, and tiny micro-motions** are umbrella factors that cover multiple **real acquisition effects**. Together, they explain why consecutive frames of a static scene can differ at the pixel level, **even when no real scene change occurs at all**.

Note (2): “**Acquisition-Related Variability**” **does not indicate real changes**, but they must be mitigated before any reliable change detection can be performed, otherwise the change map may be dominated by false positives and noise.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

(B) – Second Class – Scene and Reference Complexity

Beyond “Acquisition-Related Variability”, change detection becomes even more challenging in real-world scenarios, where the reference is **often more complex than a single frame** (i.e., a mosaic) and the observed scene may be **dynamic** or **gradually evolving over time**. Let us now focus on the key concepts introduced above:

B.1 – Camera Motion and Mosaic-Based References:

In many real-world scenarios, the camera is not fixed but moves across the scene (e.g., PTZ cameras, mobile platforms, UAVs). As a result, **the reference is often not a single frame, but a mosaic-based representation** constructed by stitching multiple overlapping RGB images, thus providing a wider and more stable view of the observed area.



Main Sources of Error in Image Mosaicking:

- **B.1.1 – Geometric Misalignment Between Frames:** Imperfect alignment causes local spatial inconsistencies.
- **B.1.2 – Accumulation of Registration Errors (Drift):** Small alignment errors accumulate across multiple frames.
- **B.1.3 – Radiometric Inconsistencies Across the Mosaic:** Brightness and color variations across mosaic regions.
- **B.1.4 – Scale Variations and Geometric Distortions:** Non-uniform scale and warping due to camera motion.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

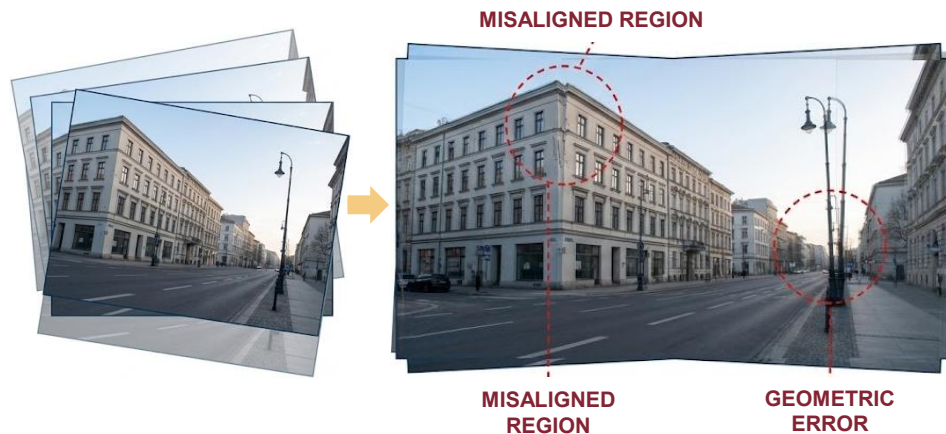
B.1.1 – Geometric Misalignment Between Frames:



It occurs when **two (or more) overlapping frames** of the same scene cannot be perfectly aligned during mosaic construction. This happens because **the alignment model cannot perfectly describe how the camera and the scene actually move**. As a result, after warping, some areas overlap correctly while others remain spatially inconsistent. Typical causes include:

- **Feature Mismatches;**
- **Rolling Shutter Effects;**
- **Viewpoint Changes / Parallax Error;**
- **Simplified Geometric Models (e.g., Homography).**

B.1.2 – Accumulation of Registration Errors (Drift):



It occurs when small residual alignment errors between consecutive overlapping frames **accumulate during mosaic construction**. Even if each misalignment is very small, **their propagation leads to a global geometric drift, thus gradually deforming the final mosaic**. Typical causes include:

- **Sequential (Pairwise) Alignment;**
- **Absence of Loop-Closure Mechanisms;**
- **Lack of Global Consistency Constraints;**
- **Long Image Sequences or Wide Camera Motion.**



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

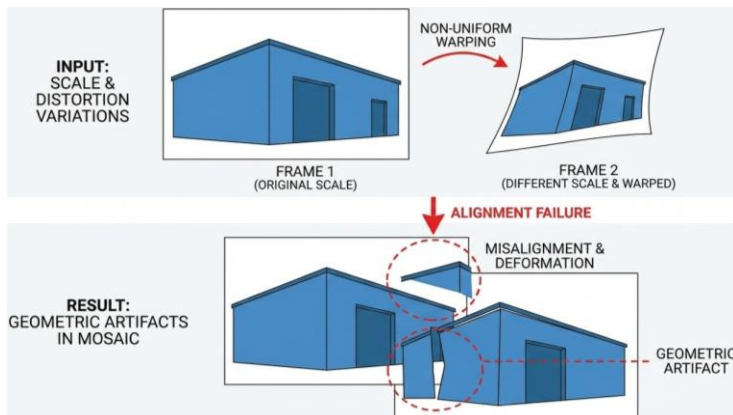
B.1.3 – Radiometric Inconsistencies Across the Mosaic:



They occur when different parts of a mosaic show **visible differences in brightness, color, or contrast, even in overlapping regions representing the same scene**. These differences arise because images are acquired under varying radiometric conditions. As a result, the mosaic may contain **visible seams or intensity discontinuities**. Typical causes include:

- **Sensor and Lens Effects;**
- **Varying Illumination Conditions;**
- **Automatic Camera Adjustments;**
- **Independent Radiometric Processing of Frames.**

B.1.4 – Scale Variations and Geometric Distortions:



They occur when frames are captured **at different scales or undergo non-uniform warping during mosaic construction**. As a result, the same scene elements may appear **larger/smaller or slightly deformed across frames**, making alignment harder and introducing geometric artifacts in the final mosaic. Typical causes include:

- **Lens Distortion and Rolling Shutter Effects;**
- **Zoom Changes in PTZ Cameras (Variable Focal Length);**
- **Simplified Geometric Models (e.g., a Single Homography);**
- **Viewpoint Changes in Non-Planar Scenes (Perspective Effects).**



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

B.2 – Dynamic Scenes and Non-Stationary Backgrounds References:

In many real-world scenarios, the observed scene is not truly static, even when the camera is fixed. **Urban environments contain moving objects** (e.g., vehicles, pedestrians), while **outdoor scenes include natural motion** (e.g., trees and grass moving with wind, water surfaces, drifting smoke). **These dynamics produce continuous and often repetitive pixel-level variations that occur within short time intervals and may affect large regions of the image.** As a result, a simple comparison between a reference frame and a current frame can generate apparent changes that do not correspond to significant events, **but rather to the normal, non-stationary behavior of the scene.** Typical examples include:

Water Surfaces with
Waves or Ripples



Moving Shadows, Reflections,
Smoke, or Clouds



Pedestrians and Vehicles
Moving in Urban Environments



Vegetation (e.g., Trees, Grass)
Swaying due to Wind



Important Note (1): The scene content changes instantaneously, but the scene itself remains the same.

Important Note (2): What counts as a “change” depends on the application context and the goal of the change detection system. For example, in restricted/quarantine-area monitoring, **grass motion** is typically **irrelevant** and should be ignored, whereas the presence of **people or vehicles** is significant and **must be detected as a change**.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

B.3 – Gradual Scene Evolution and Long-Term Changes:

In other cases, the scene does not change quickly but **evolves slowly over time, following long-term processes rather than instantaneous motion**. These changes may occur over **days, weeks, or months**, and are often subtle when observed between consecutive frames, but significant when analyzed over longer time spans. **Such evolution challenges the notion of a static reference, since the “normal” appearance of the scene itself changes over time**. If not properly modeled, gradual evolution can be mistakenly detected as anomalous or unexpected change. Typical examples include:

Vegetation Growth, Crop Cycles, or Seasonal Changes



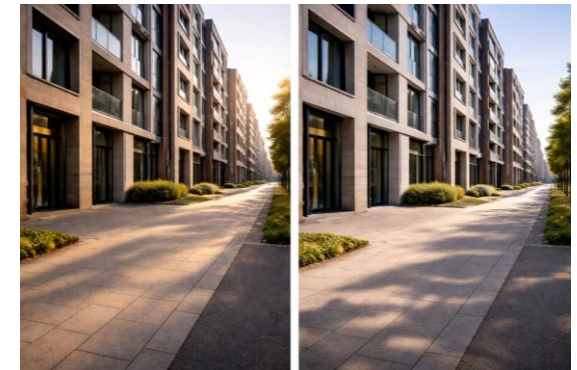
Construction Progress or Gradual Urban Development



Weathering, Erosion, or Material Degradation



Long-Term Illumination Changes Across Seasons



Important Note (1): The scene content changes **permanently and cumulatively over time**, meaning that the baseline appearance of the **environment gradually evolves** and cannot be reliably represented by a single fixed reference frame.

Important Note (2): In this setting, reliable change detection requires an **adaptive or time-aware reference**, since differences between frames may represent **expected long-term evolution rather than real changes**.



ADVANCED COMPUTER VISION

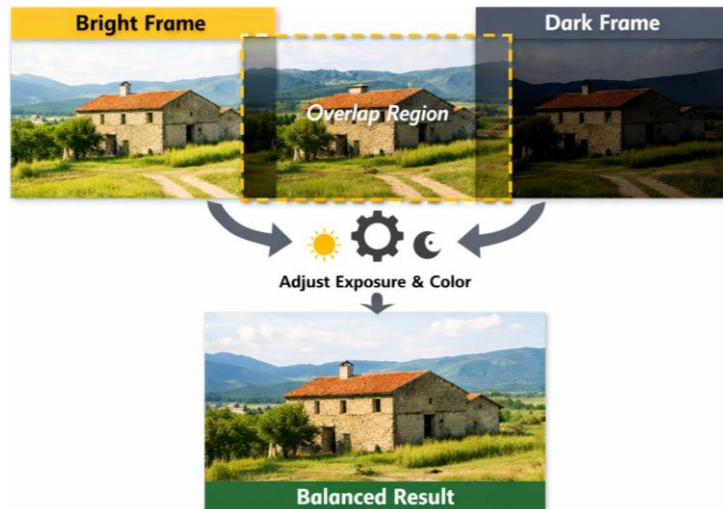
Change Detection • Saliency Detection • Anomaly Detection

Cross-Cutting Solutions for Class I and Class II Change Detection Challenges:

In recent years, the computer vision community has proposed a number of **advanced solutions** designed to address **multiple change detection challenges simultaneously**. Rather than targeting individual issues in isolation, **these approaches aim to mitigate several sources of variability and complexity in a unified manner**, covering both **acquisition-related effects** (i.e. problems seen in Class I) and **scene/reference complexity** (i.e., problems seen in Class II).

S.1 – Joint Photometric Normalization for Robust References

Core Idea: Before blending frames into a mosaic (and before performing change detection on them), **it is necessary to apply photometric compensation** so that **exposure and color differences** across **overlapping regions** do not create **visible seams** or **false differences**. Here are a few practical strategies that can be used **alone or combined** to mitigate common change detection issues.



S.1.1 – Exposure / Gain Compensation (Radiometric Equalization) in Overlap:

When consecutive frames overlap, **the same scene regions should ideally have similar intensity and color**. In practice, exposure, gain, and illumination may vary between frames. This step estimates a **global or local brightness/color correction** using the overlapping areas, so that frames match more closely before blending or change detection.

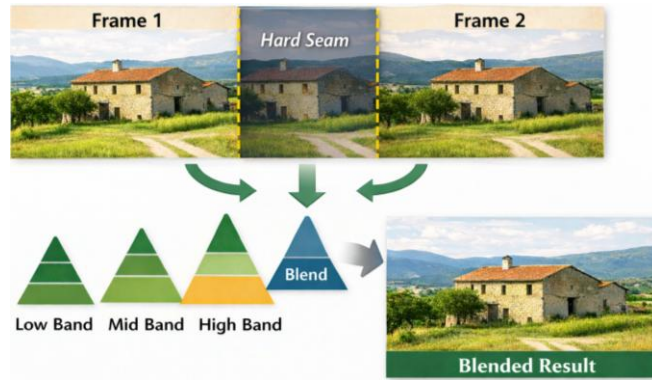
Mitigates: **A.1** (illumination variability), **B.1.3** (radiometric seams), and partially **B.2** (lighting-related effects such as shadows/reflections).



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

S.1.2 – Seam-Aware Blending (Multi-Band / Pyramid Blending)

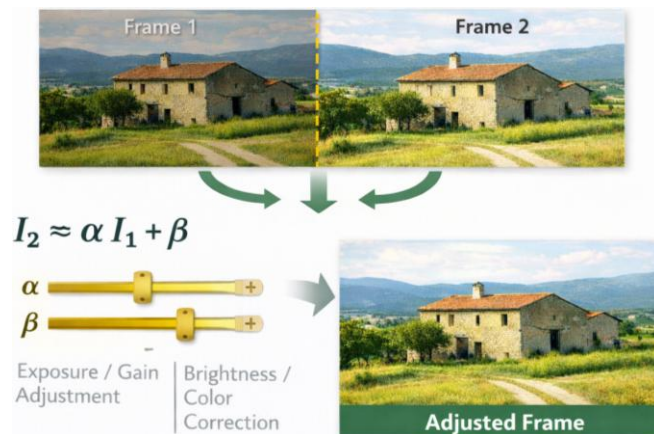


Even after photometric compensation, small differences may remain across frames. **Seam-aware blending** reduces visible discontinuities **by combining images across multiple frequency bands** (i.e., coarse-to-fine = blending first on low-frequency (i.e., global illumination), then on high-frequency details (i.e., edges/textures)), **thus producing smoother transitions at the boundaries**. This is especially effective when stitching multiple frames into a mosaic, where seams would otherwise highlight radiometric inconsistencies.

Mitigates: B.1.3 (visible seams and intensity discontinuities across the mosaic).

Note: In this context, we consider incremental (online) mosaicking, not global (offline) mosaics.

S.1.3 – Radiometric Calibration / Relative Calibration



In applications where **radiometric consistency matters** (e.g., UAV monitoring and long-term surveillance), **simple exposure or histogram equalization may not be enough**. Radiometric (relative) calibration makes the camera response more consistent across frames, reducing systematic brightness/color differences caused by the imaging pipeline (i.e., sensor, optics, and in-camera processing). **This helps reduce seams in mosaics and keeps the reference appearance stable over time.**

Mitigates: A.1 and B.1.3. Supports B.3 when the reference must remain consistent over long time spans.

Note: Radiometric (relative) calibration is an advanced step that estimates consistent brightness/color corrections to reduce systematic differences across frames, especially in long-term acquisitions.

General Note: Approaches S1.1 → S1.2 often form a natural sequence in image stitching pipelines. Approach S1.3 can be applied beforehand or used to support Approach S1.1, especially in challenging or long-term scenarios.



ADVANCED COMPUTER VISION

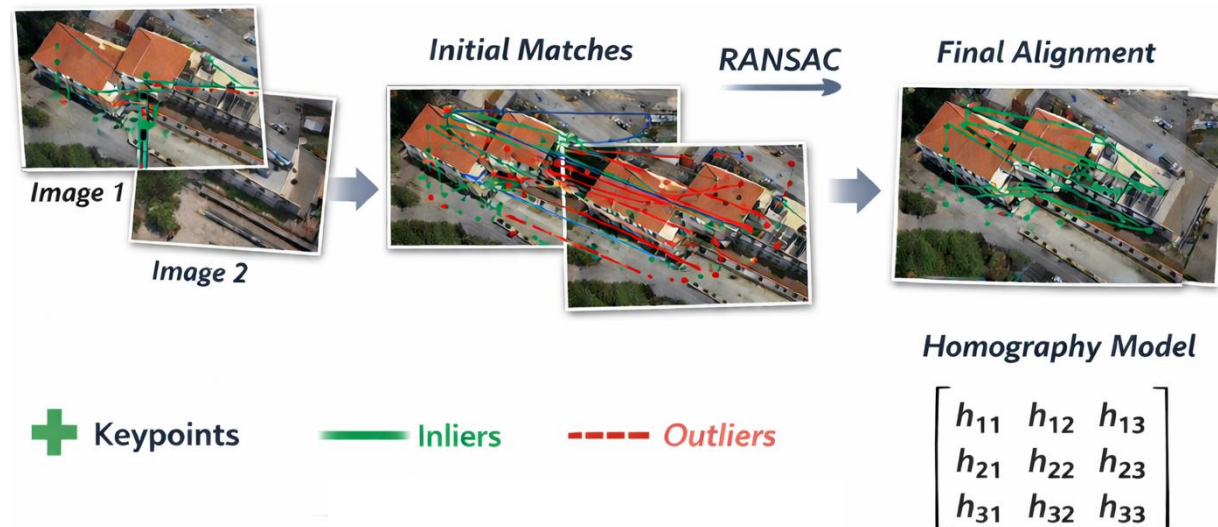
Change Detection • Saliency Detection • Anomaly Detection

S.2 – Robust (and Learned) Registration to Stabilize the Mosaics

Core Idea: Improving **image registration** (i.e., alignment of frames to the same coordinate system) makes mosaics more **stable and reliable** by reducing **geometric misalignment**, **limiting drift**, and **mitigating distortions**, especially when camera motion is present. Here selected strategies that can be used **alone or combined** to mitigate common change detection issues caused by **misalignment**.

S.2.1 – Feature-Based Alignment + Robust Estimation (e.g., RANSAC)

This is the classic and still widely used approach for registering overlapping RGB frames. **Keypoints** (e.g., SIFT/ORB/AKAZE) are detected in both images and matched across the overlap region. Since some matches are **wrong (outliers)**, a robust estimator such as RANSAC is used to keep only consistent correspondences and estimate a **geometric model** (e.g., **affine or homography**). This step greatly improves alignment reliability under **noise**, **small viewpoint changes**, and **local ambiguities**, and is particularly effective for PTZ and classical mosaicking pipelines.



Mitigates: **A.3** (tiny micro-motions → misalignment), **B.1.1** (geometric misalignment between frames).

Note: RANSAC helps **ignore wrong matches** and **estimate a reliable geometric transformation** using only consistent feature correspondences.



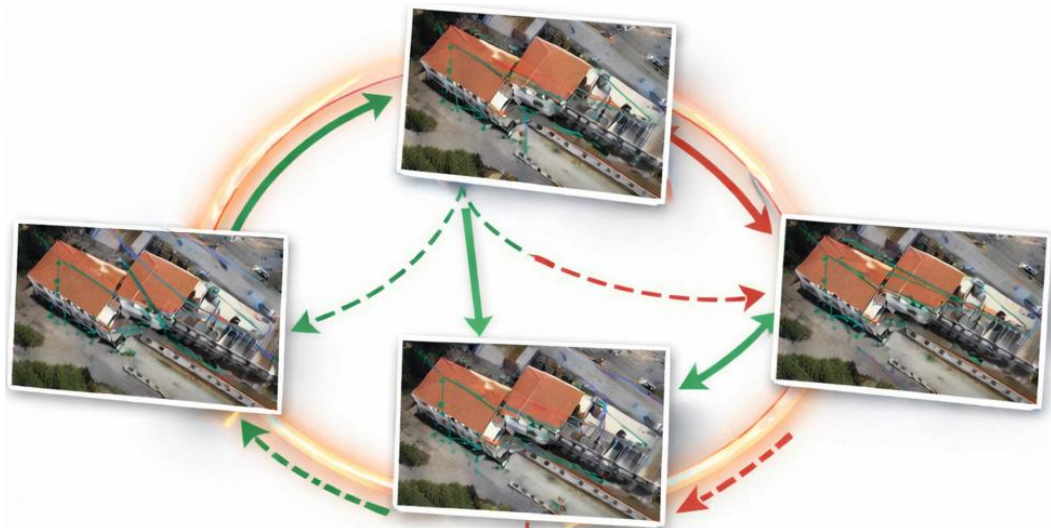
ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

S.2.2 – Global Consistency and Drift Control (Global Optimization / Loop Constraints)

When a mosaic is built from many frames, **aligning each frame only to the previous one can accumulate small errors over time**. These small errors add up and cause drift, slowly deforming the mosaic. To reduce this effect, **global consistency methods optimize the alignments across many frames together** (either the whole sequence (**offline**) or a sliding window (**online**)).

Global consistency methods can also use **loop constraints**: **when the camera revisits an area, the system enforces agreement with the already-stitched region to correct the drift**.



In Other Words:

- **Why Pairwise Alignment is not Enough:** Each alignment step is almost correct, but not perfect. Over many frames, tiny errors accumulate and the mosaic gradually “slides” away from the true geometry (drift).
- **What Global Consistency Does:** Instead of trusting only local pairwise matches, it adjusts transformations across many frames together, so the full mosaic remains coherent overall.
- **What Loop Closure Means (Intuition):** If the camera sees a previously observed region again, the system adds a constraint: the new view must match the old area. This helps correct the accumulated drift and “close the loop” cleanly.

Mitigates: B.1.2 (accumulation of registration errors / drift).

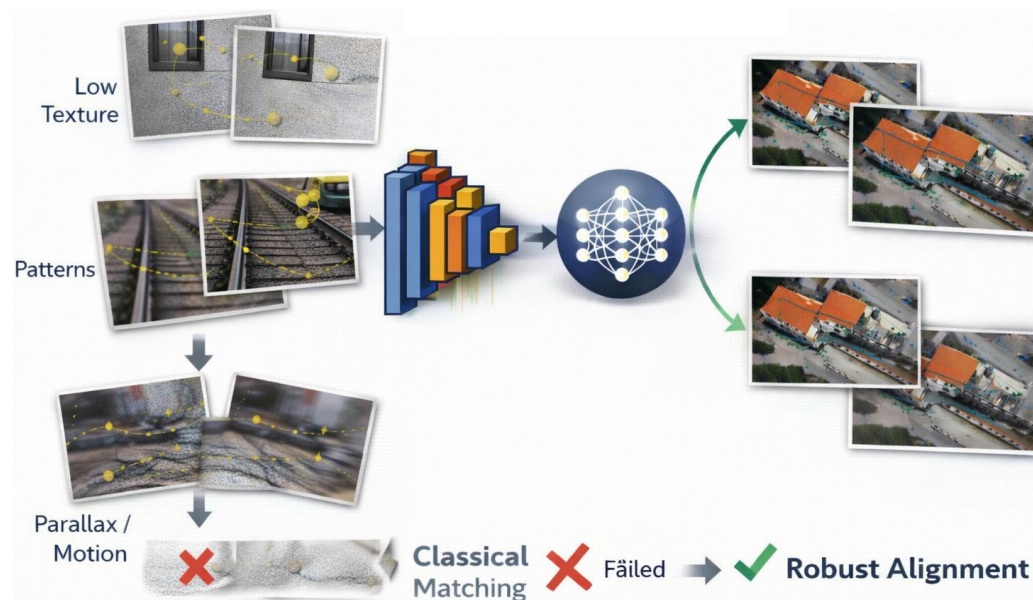


ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

S.2.3 – Deep Homography / Learned Alignment (Modern Registration Models)

More recent approaches replace or complement traditional feature-based pipelines with **learned models** that estimate the alignment directly. Deep alignment networks can be more robust when classical matching becomes unreliable (e.g., low texture, repetitive patterns, illumination changes, partial occlusions). These methods can learn to predict transformations that better handle situations such as mild parallax, imperfect feature matches, or complex motion that is not fully captured by a simple model. **In practice, they are often used as a stronger “alignment module” that improves downstream mosaicking stability.**



In Other Words:

- **Learned Alignment Instead of Only Keypoint Matching:** Modern methods use deep models to predict the geometric alignment directly, or to support traditional feature-based pipelines.
- **More Robust in Difficult Conditions:** Deep alignment networks often work better when classical matching struggles (e.g., low texture, repetitive patterns, illumination changes, partial occlusions).
- **Handles Harder Motion and Improves Mosaics:** They can estimate transformations more reliably under mild parallax, imperfect matches, or complex motion, acting as a stronger alignment module that stabilizes mosaicking and reduces downstream errors.

Mitigates: B.1.1 (misalignment), partially B.1.4 (scale variations / distortions), and can indirectly reduce drift by improving each registration step.



ADVANCED COMPUTER VISION

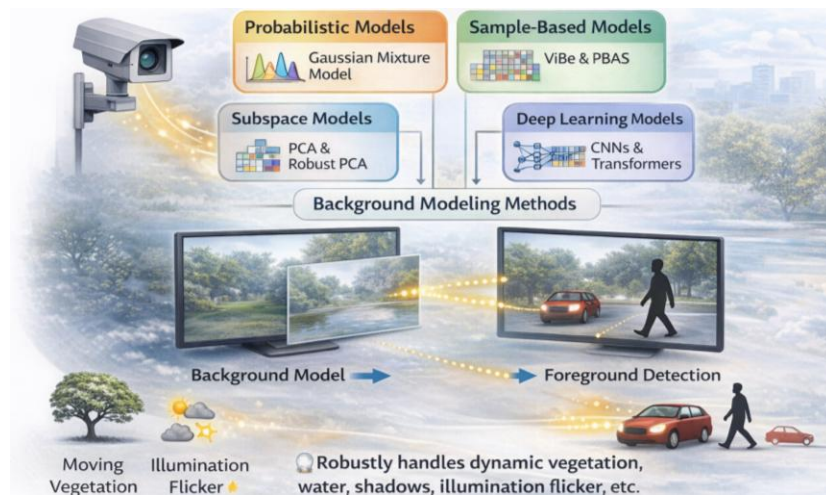
Change Detection • Saliency Detection • Anomaly Detection

S.3 – Adaptive Reference Modeling for Dynamic & Evolving Scenes

Core Idea: rather than comparing each new frame to one fixed “snapshot” (i.e. **reference frame or reference mosaic**) of the scene, we maintain a **reference model** that represents what the scene usually looks like. **This model is updated over time**, so it can follow normal background variations (e.g., periodic lighting shifts, moving trees, rippling water) **without treating them as real changes**. In this **way system focuses on detecting significant / real foreground events**, while the reference gradually adapts to the normal, expected behavior of the environment (*Note: background variations must be defined: lighting shifts within a day? Over a year?, People?, Vehicles?*).

S.3.1 – Background Modeling (Classical but Still State of the Art in Surveillance)

Classical background modeling methods build a reference model of what is normally expected at each pixel or region (using, e.g., probabilistic distributions or sample-based models), so the system can separate **normal background dynamics from actual foreground events**. Thanks to their efficiency, stability, and ease of deployment, **these approaches are still widely used and effective in surveillance applications**.



Simple Example: A basic (i.e., didactic) adaptive background can be implemented as a **running average** per pixel: $B_t = (1 - \alpha) B_{t-1} + \alpha I_t$, where:

- $B_t \rightarrow$ estimated background at time t ;
- $B_{t-1} \rightarrow$ background estimate at the previous time step;
- $I_t \rightarrow$ current image acquired by the sensor;
- $\alpha \rightarrow$ learning rate (controls how fast the background is updated).

Mitigates: B.2 (dynamic scenes and non-stationary backgrounds).



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

S.3.1 – Background Modeling (Classical but Still State of the Art in Surveillance)

Further Information on the Example:

$$B_t = (1 - \alpha)B_{t-1} + \alpha I_t$$

What the Terms Represent:

- B_t : Estimated background at time t
- B_{t-1} : Background estimate at the previous time step
- I_t : Current image from the sensor
- α : Learning rate (controls update speed)

How It Works (Intuition)

Previous Background "What it believed before" → **Blended Update** "Weighted Average" → Current Pixel "What it sees now"



Given: $B_{t-1} = 100$, $I_t = 130$
 $\alpha = 0.05$
 $B_t = 0.95 \times 100 + 0.05 \times 130 = 101.5$
→ Moves Slowly → Not Jumps to 130!

Numerical Example

- $B_{t-1} = 100$, $I_t = 130$, $\alpha = 0.05$
- $B_t = 0.95 \times 100 + 0.05 \times 130 = 101.5$

Case 1: α Small (e.g., 0.01)

- Slow Adaptation
- Person not absorbed



Case 2: α Large (e.g., 0.1)

- Fast Adaptation
- Car becomes background



Why Useful for Change Detection

- Temporary Objects Ignored
- Slow Lighting Changes Absorbed
- Sudden Events Detected



Key Limitation: Can't handle complex or persistent backgrounds



How It Works in Practice (Intuition)

Each background pixel is updated as a **weighted average** between:

- What the background believed before;
- What the pixel shows now.

Case 1 - Small α (e.g., 0.01)

- The background updates **slowly**;
- A new intensity needs **many frames** to be absorbed.

Good for:

- Slowly changing illumination;
- Preventing people or cars from becoming background.

Example:

- A person appears for 2-3 frames → **not absorbed**;
- Sunlight changes over 30 minutes → **absorbed**.

Case 2 - Large α (e.g., 0.1)

- The background updates **quickly**;
- Persistent objects are absorbed **rapidly**.

Example:

- A parked car → becomes background after a short time;
- Useful for scenes that change frequently.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

S.3.2 – Selective Update / Forgetting Factor (Controlled Reference Update)

Core Idea: To handle gradual scene evolution, the reference model must be updated carefully. Selective update strategies introduce a **forgetting factor** or **controlled learning rate**, allowing the reference to adapt slowly over time **while avoiding the immediate absorption of relevant or anomalous events**. This balance helps preserve long-term stability while remaining responsive to genuine, permanent changes in the scene, even when conditions shift subtly from day to day.



Mitigates: B.3 (gradual scene evolution) and partially B.2.

In other words:

- **Start From a Reference Model:** Maintain a background/reference that represents the scene under “**normal**” conditions.
- **Update Slowly Using a Controlled Rate:** Apply a small learning rate (or a forgetting factor) so the **background/reference changes gradually**, not frame by frame.
- **Update Only When It Is Safe (Selective Update):** Update the model mainly for pixels/regions classified as **background**, and avoid updating areas that look like **relevant events**.
- **Balance Stability and Adaptation Over Time:** The reference stays stable in the short term (events are not absorbed immediately), but it still adapts to **true long-term changes** as they persist.

Note: The running average (i.e. example in S.3.1) is a simple example of background update, while selective update strategies define **when and how** any reference model should adapt over time.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

S.3.3 – Multi-Model or Context-Aware Reference Modeling

Core Idea: In many real-world scenarios, a single evolving reference **is not sufficient** because the same scene can exhibit multiple normal states depending on the context (e.g., day vs. night, different weather conditions, varying activity levels). **Multi-model or context-aware reference modeling** addresses this by maintaining multiple reference or background models, each representing a different typical state of the scene. At each time, the system **selects, switches, or softly combines** the most appropriate model based on the current observations or contextual cues, rather than forcing all variations into a single reference.



Mitigates: B.2 (dynamic scenes with recurring conditions).

Mitigates: B.3 (long-term changes with distinct regimes).

In other words:

- **Multiple Normal States of the Same Scene:** The same location may have different “normal” appearances (e.g., day vs night, dry vs rainy, working hours vs night hours). A single reference is often insufficient.
- **Context Selection Instead of Continuous Adaptation:** Rather than slowly updating one reference, the system selects the most appropriate model based on context cues (e.g., illumination level, scene statistics).
- **Avoids Contaminating the Reference:** Sudden but recurring conditions (e.g., headlights at night, rain reflections) are handled by switching models, not by forcing the reference to adapt.
- **Typical Examples:** Daytime background vs nighttime background; Dry road vs wet road; Summer illumination vs winter illumination; Empty street vs regularly parked vehicles.

Note: Complementary to Selective Update: Selective update adapts within one model; multi-model strategies switch between models.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

A Brief Recap: Building the Reference Model

So far, we have focused on how to build a reliable reference model. In particular, we have introduced the **main classes of challenges** that affect reference construction in real-world change detection:

- Acquisition-Related Variability (**A.1-A.3**);
- Scene and Reference Complexity (**B.1-B.3**).

We then discussed **cross-cutting classes of solutions** designed to mitigate these issues:

- Joint Photometric Normalization for Robust References (**S.1**);
- Robust (and Learned) Registration to Stabilize the Mosaics (**S.2**);
- Adaptive Reference Modeling for Dynamic & Evolving Scenes (**S.3**).



Note: Together, these techniques aim to produce a reference or background model that is **as robust, stable, and coherent as possible under real operating conditions**, for example:

- Day-night appearance shifts over 24 hours (lighting, shadows, exposure);
- Seasonal variations across months (sun angle, typical illumination patterns);
- Gradual scene evolution such as moderate structural changes (wear, slow environmental effects);
- Recurring “normal” dynamics like waving vegetation, water ripples, or periodic traffic patterns.

In short, the goal is a reference that stays consistent enough for comparison, while still adapting to expected, context-dependent variations of the same scene.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

An Important Clarification

The solution classes introduced so far (**S.1-S.3**) are **general strategies**, not fixed pipelines. They provide conceptual tools to address multiple problems at once, but **the actual reference model is always application-dependent**. In practice, how a reference/background is built depends on several factors, including:

- **The environment** (e.g., indoor vs. outdoor, urban vs. natural scenes);
- **The task** (e.g., generic surveillance, intrusion detection, monitoring, safety, traffic analysis);
- **What is considered “background” versus “relevant change”**.

This means that the same scene can be modeled in different ways depending on the application. For example:

- Vegetation may be modeled as dynamic background;
- Static structures (e.g., walls, benches, railings) are typically treated as stable reference;
- Objects such as vehicles or people may be:
 - Excluded from the reference (e.g., intrusion detection), or
 - Incorporated into it (e.g., traffic or parking monitoring).

In other words: There is no single “correct” reference model: it is the result of design choices driven by context, scene characteristics, and system objectives.



ADVANCED COMPUTER VISION

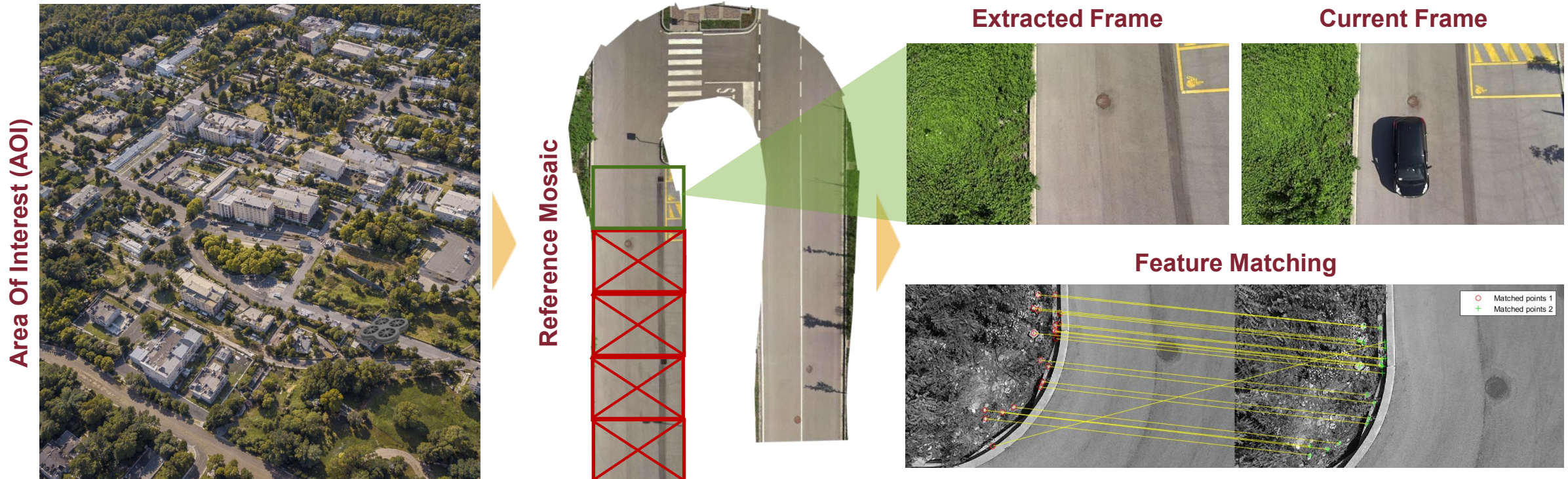
Change Detection • Saliency Detection • Anomaly Detection

Final Steps of Change Detection

Up to now, we have focused on how to construct a **reliable and meaningful** reference / background model. However, building the reference is only the first part: change detection also requires **(1) aligning** and **(2) comparing** new observations.

(1) – Alignment of the Current Observation to the Reference

Before computing differences, the current frame must be geometrically aligned to the reference model (a single frame, a mosaic, or a background model), so that **the same scene elements correspond to the same spatial locations**. That is:





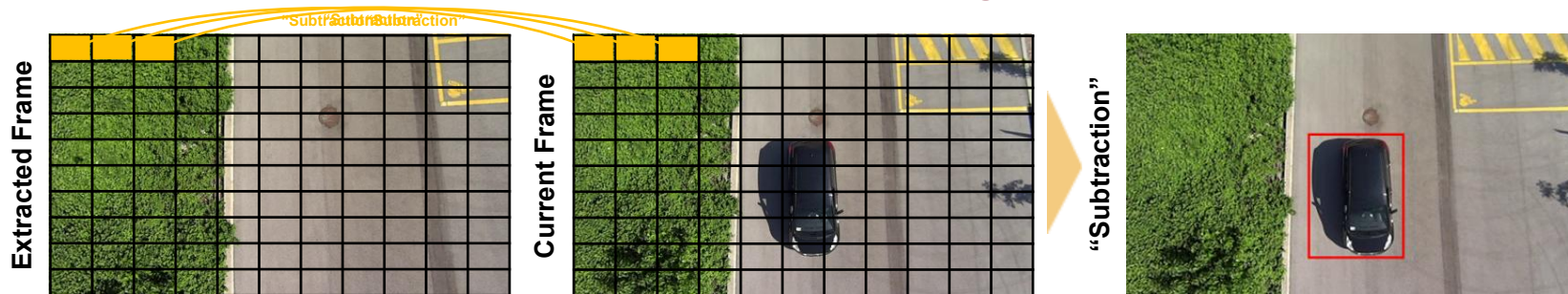
ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

(2) – Robust Change Computation

Once the current frame is aligned to the corresponding part of the reference model, changes are usually **not** computed by simple pixel-wise subtraction (except in **very simple settings**, such as fixed cameras under controlled conditions). Instead, differences are evaluated at the patch level, either **as scalar scores** or **as descriptor embeddings**, using robust measures that can tolerate **noise**, **residual misalignment**, and **natural variability**. Only by combining proper alignment with robust comparison strategies **real changes** can be distinguished from **artifacts**.

Example of Patch-Based Scalar Score Differencing (Handcrafted)



In other words: differences are evaluated at the **patch level** rather than at individual pixels, by computing one (or a small set of) **scalar change scores** for each patch. This local aggregation makes the comparison more robust, helping isolate meaningful changes even in the presence of noise and small residual misalignments.

Examples of Patch-Based Scalar Scores

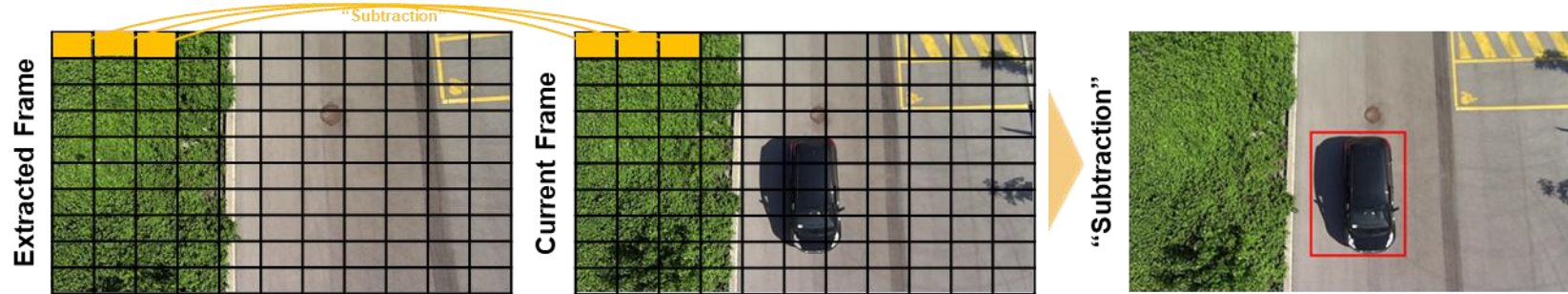
- **Pixel-Statistics / Photometric Scores:** Mean / Median Difference, Variance / Standard Deviation Change, Mean Squared Error / Root Mean Squared Error, Mean Absolute Error / Mean Absolute Deviation, Z-Score / Normalized Difference, Trimmed Mean Difference, Percentile Shifts, Huber-Style Loss, etc...
- **Distribution Scores:** Histogram Distances (e.g., χ^2 Distance, Bhattacharyya / Hellinger Distance, Jensen-Shannon Divergence, Earth Mover's Distance / Wasserstein-1), Kolmogorov-Smirnov Statistic, Mutual Information Drop (Decrease in Mutual Information as an Indicator of Local Inconsistency, etc...
- **Texture / Structure as Statistics:** GLCM Statistics (e.g., Contrast, Homogeneity, Energy), Local Variance / Local Energy, Edge Density / Edge Count, Gradient Magnitude Summary (e.g., Mean / Variance), Orientation Consistency (Dispersion of Orientations), Structural Similarity Index Measure, etc...
- **Binary / Morphological Scores:** Area / Occupancy Change, Shape Compactness / Eccentricity Change, Connected Components Change, Boundary Length / Perimeter Change, Convex Hull Area/Solidity Change, Centroid Shift, Euler Number Change, Orientation Change, Skeleton Length Change, etc...



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

Example of Patch-Based Descriptor Embedding Comparison (Handcrafted)



In other words: differences are evaluated by comparing **descriptor embeddings extracted from corresponding patches**, rather than relying on a small set of scalar patch-level scores. This representation-level comparison is often more robust to noise, illumination fluctuations, and small residual misalignments.

Examples of Patch-Based Descriptors Embedding

- **Gradient / Shape Descriptors (Dense Vectors):** HOG (Histogram of Oriented Gradients) Vectors, Dense SIFT / SIFT-Like Descriptors, Dense SURF / SURF-Like Descriptors, Dense AKAZE / AKAZE-Like Descriptor, Dense ORB / ORB-Like Descriptors, DAISY Descriptors (Designed for Dense Matching), etc...
- **Texture Descriptors (Dense Vectors):** LBP (Local Binary Pattern) Histograms (Computed Locally), Gabor Filter Bank Responses (Energies Across Scales / Orientations as a Vector), Wavelet / Multi-Resolution Texture Features, Co-Occurrence-Based Descriptor Vectors (stacked GLCM statistics per direction), etc...
- **Color / Photometric Descriptors (Vectors):** Color Histograms (RGB/HSV/Lab), Color Correlograms (Spatial Correlation of Colors), Color Moments (Mean/Standard Deviation/Skewness per Channel Concatenated), Chromaticity Histograms (More Illumination-Tolerant), Local Color Covariance Descriptors, etc...
- **Aggregated Local Descriptors (Vectors):** Bag of Visual Words / Vector of Locally Aggregated Descriptors / Fisher Vector (Computed Locally), Scattering Transform Descriptors, Local Gaussian Mixture Model / Mixture-of-Gaussians (MoG) Statistics over Local Descriptors (e.g., mixture means/variances), etc...

Note (Summarizing)

In handcrafted patch-based differencing, we can either **(a)** summarize each patch pair with **scalar scores** computed from simple measures (e.g., photometric/texture/distribution measures, and so on) and detect change by comparing these scores across reference versus current patches, or **(b)** encode each patch into a **descriptor embedding** built from one or multiple handcrafted descriptors (e.g., HOG, LBP histograms, DAISY/dense SIFT, BoVW/VLAD, and so on) and detect change by measuring distances between embeddings in descriptor space.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

Patch-Based Scalar Scores vs Patch-Based Descriptor Embeddings (1/2)

What is Compared (Representation)

- **Patch-Based Scalar Scores:** each patch pair is summarized by **one (or a small set of) scalar scores**; change is inferred from **score discrepancies** between corresponding patches;
- **Patch-Based Descriptor Embeddings:** each patch is represented by a **descriptor vector (embedding)**; change is inferred from **embedding discrepancies (i.e., distances)** between corresponding patches.

What is User-Defined (Handcrafted Setting)

In both handcrafted approaches, we choose:

- **The Spatial Partition** (i.e., patch size, stride, overlap, grid layout);
- **The Comparison Design:** which scalar scores (and scalar-score differencing) and which descriptors/embeddings (and embedding comparison) and the associated distance metric.

How Change is Computed (Scores vs Distances)

- **Scalar Score Differencing:** change is detected via **direct patch-level scores** (e.g., differences/ratios/normalized scores, distribution tests or histogram distances), possibly combined across multiple measures;
- **Descriptor Embedding Comparison:** change is detected via a **distance between patch embeddings** (e.g., L1/L2/cosine for continuous descriptors; χ^2 /Bhattacharyya/EMD for histogram-shaped descriptors), optionally after feature fusion.

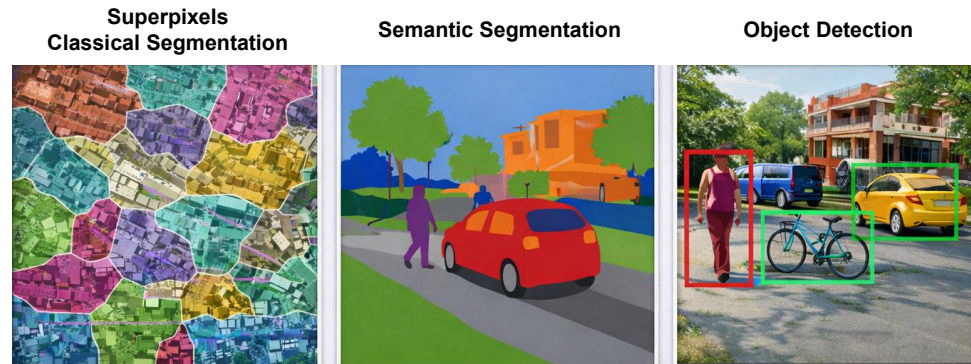


ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

Patch-Based Scalar Scores vs Patch-Based Descriptor Embeddings (2/2)

Beyond a Regular Symmetric Grid (Still User-Designed, but Content-Adaptive)



- The spatial support does not have to be a square, fixed grid: it can be defined by **irregular, content-adaptive regions** (region-based/object-based comparison).
- Regions can be obtained via **classical segmentation** or **superpixels**, or via higher-level partitions such as **semantic segmentation** masks and **object detections**.
- The partitioning strategy and its parameters remain user-chosen, even though the resulting regions are **data-driven** and **adapt** to the scene content.

Note: **Superpixels** partition an image into small, **homogeneous regions** based on pixel similarity and local coherence (i.e., focus on pixel-level similarity), whereas **classical segmentation** aims to directly divide the image into **meaningful regions or objects** according to the task (i.e., focus on object-level or task-level segmentation).

Learned Representations (Moving Beyond Handcrafted)



With **CNNs/ViTs**, patch representations are produced by a **learned encoder** rather than by manually selecting handcrafted descriptors.

- **CNNs:** an input image is transformed into a stack of **feature maps**. Each spatial location in the feature maps corresponds to a local region in the original image and provides an **embedding vector** (the channel values). The spatial resolution of this grid is set by the network's **stride/downsampling**.
- **ViTs:** the image is explicitly split into **fixed-size patches** (a user-chosen patch size). Each patch is mapped to a **token embedding**, and the transformer refines these embeddings through self-attention.

Note: You still choose some hyperparameters (e.g., patch size, stride, embedding dimension), but the feature extractor and the embeddings **are learned from data during training**.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

A Case Study: A UAV Video Dataset for Mosaicking and Change Detection From Low-Altitude Flights (1/3)



■ Research Topic and Purpose

A. Context & Motivation:

Low-altitude UAV monitoring often requires **mosaicking** (to build a wide reference view) and **change detection** (to detect differences over time), but low-altitude flights introduce additional challenges (e.g., parallax, attitude changes, illumination/shadows).

B. Problem

Most available datasets **are not made** for low-altitude UAV mosaicking and change detection, and they often do not provide **the same flight paths recorded more than once** together with the **extra data** (e.g., GPS/telemetry) needed for fair and consistent comparisons.

C. Contributions

The paper delivers: **(1) the UMCD dataset** for low-altitude mosaicking & change detection, **(2) a baseline mosaicking pipeline**, and **(3) a baseline change-detection pipeline** to serve as a shared reference for future comparisons.

D. Benchmark Protocol & Release

The dataset includes acquisitions along similar trajectories **with/without inserted objects/persons/vehicles** across multiple environments, and the authors provide **benchmark mosaics plus evaluation metrics** for both mosaicking quality and change-detection performance.



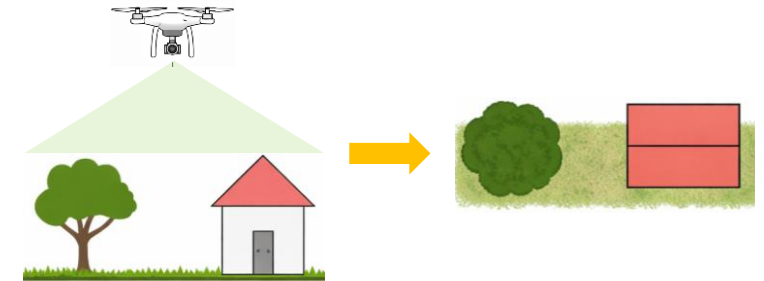
ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

A Case Study: A UAV Video Dataset for Mosaicking and Change Detection From Low-Altitude Flights (2/3)

Some Main Design Choices

Nadir View:



- **Acquisition:** 6-15m altitude, 2-12m/s speed.
- **Frame Rate:** 24-50 fps.
- **Frame Selection (Mosaicking):** keep only the minimum frames needed (speed + overlap).
- **Geo-Referencing:** GPS at frame barycenter; interpolate if GPS rate is lower (e.g., 1Hz).
- **Field Protocol:** 1 mosaic/hour (~5 min build).

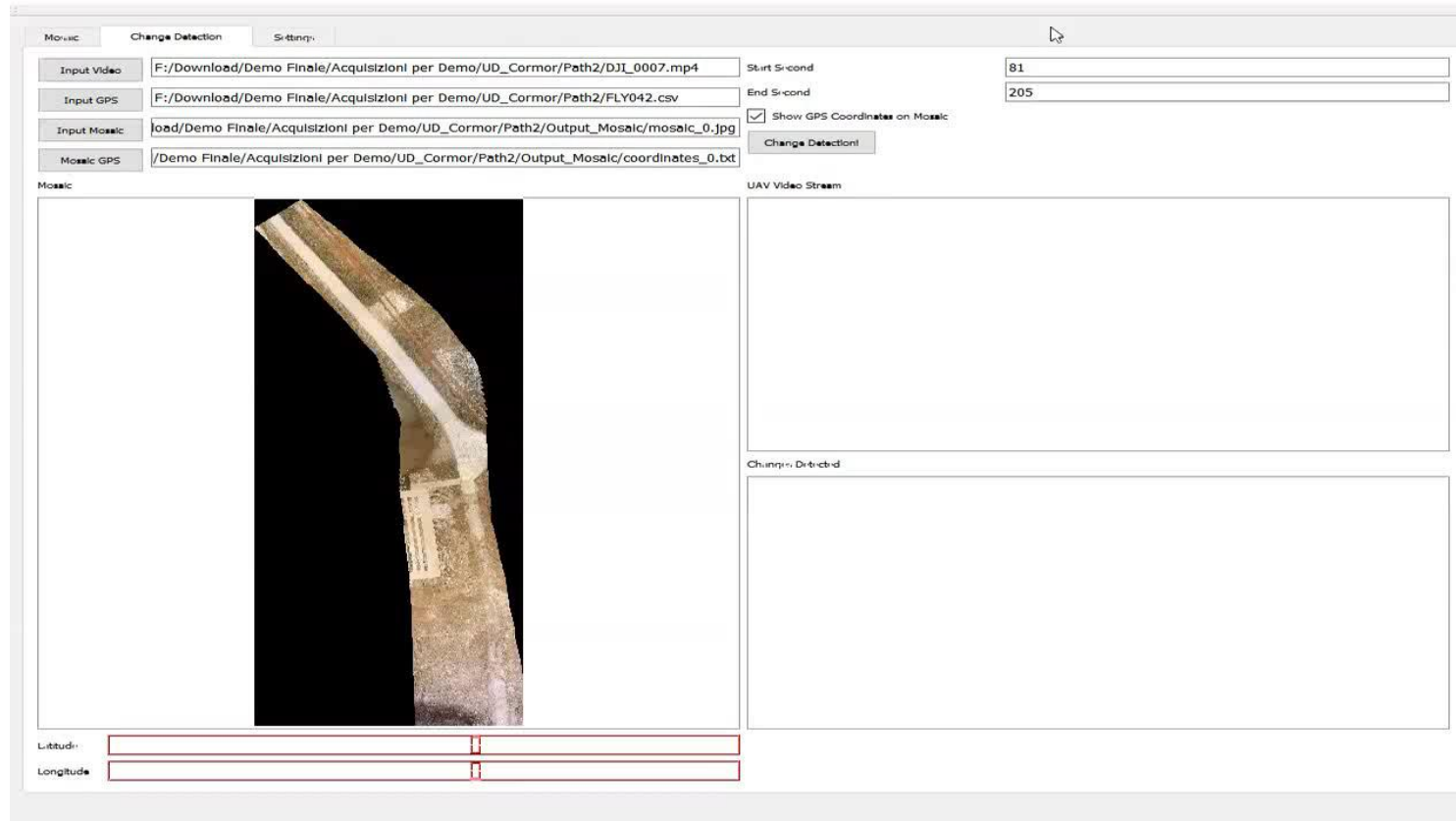
Note: A **nadir view** means the camera is pointed **straight down** at the ground, along the **vertical line** from the drone to the scene. This gives a **top-down** image, usually with **less side view** and **less distortion** than a **tilted** (oblique) view.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

A Case Study: A UAV Video Dataset for Mosaicking and Change Detection From Low-Altitude Flights (3/3)



- **Some Main Design Choices**
- **Reconnaissance Flights.**
- **During Each Reconnaissance Flight**
- **Reference Mosaic (Baseline):** considering the geo-referenced mosaic from the flight with no objects/people/vehicles (previous step).
- **Controlled Changes (Test Setup):** introduce objects/people/vehicles and repeat the flight to create “changed” observations.
- **Coarse Alignment (GPS Prior):** use the frame GPS to pick the most likely mosaic region/patch to compare.
- **Fine Alignment (Image Registration):** refine with feature-based matching to obtain pixel-level frame-patch alignment.
- **Change Detection (Comparison):** compare the current frame with the aligned mosaic patch to detect significant changes.

Note: In the **55 minutes** after the **reference mosaic** is created, the drone will perform **one or more random flights** over the entire area (or selected parts of it) to check for any **significant changes**. Then a new reference mosaic is generated, and the cycle repeats.



ADVANCED COMPUTER VISION

Change Detection • Saliency Detection • Anomaly Detection

Thesis and Collaborations Opportunities at **Prometheus Lab** (BSc / MSc / PhD) - **Change Detection**



Background Modeling and Foreground Detection for robust change maps in real indoor/outdoor scenes.



Fine Alignment and Local Subtraction to compare reference mosaics and current frames with fewer errors and false positives.



Mosaic Artifact Reduction (e.g., ghosting, seams, blur) to improve the quality of the reference model used for change detection tasks.



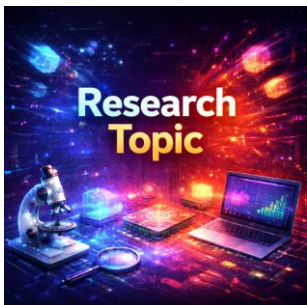
Patch-based Change Scoring to design and test scalar scores or feature descriptor embeddings under noise and small misalignment.



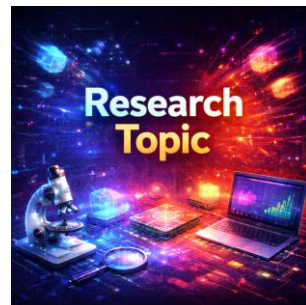
Content-Adaptive Regions to perform change detection using superpixels-based or segmentation-based regions instead of a fixed grid.



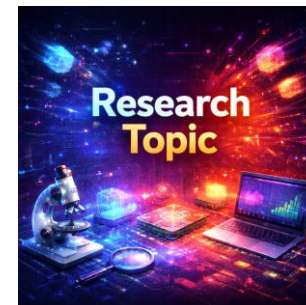
Learning-based Change Detection: in order to develop CNN/ViT embeddings for matching and change scoring, with lightweight models for UAV use (e.g. pruning).



Uncertainty and Confidence Maps to detect and filter unreliable changes caused by parallax, shadows, or moving vegetation/water.



Benchmarking on UMCD (or other Dataset) to build an evaluation pipeline (e.g., metrics, ablations, speed/accuracy trade-offs) for UAV mosaicking + change detection.



- and many other tasks (including new models and new paradigms);
- Have your own idea to propose and explore?